Programming Assignment 2: Ironbow

Due Date: October 9th, 2026, 11:59 AM

"What the eye cannot see, the pixels can reveal."- Anonymous

Objective

False-color palettes like ironbow are the standard way to display single-channel sensor data in thermal imaging. Firefighters, building and electrical inspectors, search-and-rescue teams, and medical thermography systems use them to make small intensity differences visible to the human eye.

The purpose of this lab is to convert RGBA images to grayscale and then to a false-color "*ironbow*" representation using 2D CUDA kernels on the GPU. Each pixel is processed independently by one thread. The kernels compute a luminosity-weighted grayscale intensity, map that intensity through a piecewise palette that runs from black through blue and purple to red, yellow, and white, and write the result as RGBA.

Procedure

Step 1: Copy Files: *Copy the template code to your account in AHPCC*
Step 2: Edit Files *The template consists of TODOs. Complete the TODO*
Step 3: Compile and Run the Code with *nsys_easy*
Step 4: *Convert the raw image to png, and complete the assignment in Blackboard*

Sample Output

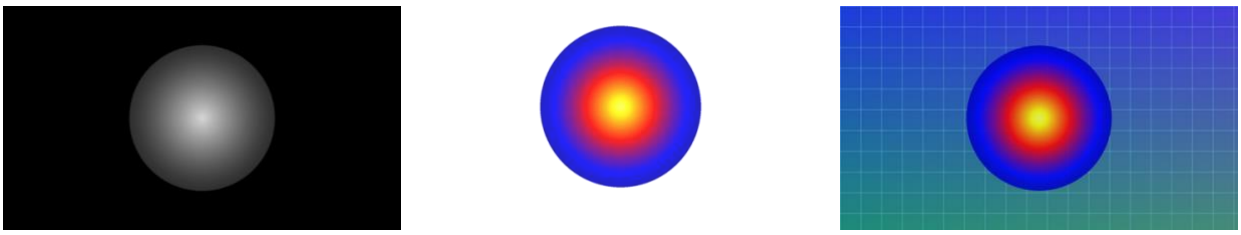


Figure 1:Grayscale and Ironbow palette of input images with and without Alpha blending

(NOTE: Your images might be different than the above sample)

Submission

1. Demo: Schedule a time with the instructor and complete the demo DemoScheduler.xlsx
2. Upload the completed code and answer questions in the Blackboard